

Proposed Use of Chatbots in Mental Health Support: Exploring Efficacy and Impact on Psychological Distress

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Abstract

Sharing personal information with a chatbot that mimics human conversation can yield positive emotional, relational, and psychological effects. However, the outcomes of such interactions, whether compromised, improved, or equivalent to human-to-human dialogue, require further investigation. This study aims to contribute to the understanding of **chatbot-assisted support for individuals experiencing psychological distress**, highlighting the need for more research and evidence on its efficacy. The potential of chatbots to act as a supportive friend to patients is a promising avenue for exploration. To achieve this, this study will employ a qualitative methodology, relying on a questionnaire to gather primary data on individuals' perceptions and experiences regarding chatbots and their potential role as a mental health support tool. The study aims to provide deeper insight into user-robot interaction, emphasizing the need for further research to establish efficacy and develop practical applications.

Keywords: chatbots, mental health, psychological distress, self-disclosure, qualitative research, support

Introduction

The advent of emerging technologies like chatbots holds immense potential to revolutionize mental health care, offering efficient and evidence-based **support mechanisms**. However, due to the novelty of this technology, there is a lack of comprehensive understanding regarding its applications, qualities, and efficacy. This paper aims to shed light on the potential of chatbots, particularly in the context of positive psychology interventions, to reach a wide audience of young adults and alleviate psychosocial distress related to health adversity.

Chatbots are broadly categorized into task-oriented and conversational types. Task-oriented chatbots operate under a predefined set of rules (Siddig & Hines,

2019). In contrast, conversational chatbots endeavor to simulate genuine human conversation, often leveraging underlying Artificial Intelligence (AI) and Natural Language Processing (NLP) technologies (Deloitte Artificial Intelligence, 2018). The application of chatbots to aid in **supporting and potentially preventing** psychological distress is a burgeoning field. The central concept involves developing a chatbot designed to serve as a supportive companion, capable of reminding patients about medication schedules and appointments with psychologists. Furthermore, the act of disclosing personal information, whether to a human friend or a chatbot, can potentially alleviate mental stress (Ho et al., 2018.). For this proposed chatbot, its knowledge base would be derived from certified academic journals. Additionally, it would incorporate a reminder system for appointments and medication and include a review section to facilitate user feedback and continuous improvement.

Psychology is defined as the scientific study of the human mind and its functions, particularly those influencing behavior within specific contexts (Oxford Languages, 1989). Consequently, a psychologist must comprehend a patient's thought processes and mental functions to effectively **support** them. Given that psychologists are not perpetually available, a chatbot offers constant accessibility, enabling patients to receive **consistent support** at any time. Research indicates that adolescents and young adults are particularly vulnerable to experiencing depression, heightened anxiety, distress, and post-traumatic stress disorder during the initial 12 to 24 months post-treatment, highlighting significant unmet mental health needs in this demographic (Kwak et al., 2013; Michel et al., 2010; National Cancer Institute, 2018). This study seeks to contribute to the ongoing development of chatbots tailored to **support individuals with** psychological conditions. It further aims to provide an in-depth examination of the interaction between the patient and the chatbot, which serves as a **supportive companion**. The significance of this research lies in its contribution to advancing AI applications that enhance quality of life.

Literature Review

Chatbots

Chatbots are fundamentally built upon question-and-answer (Q&A) systems that retrieve pertinent information from knowledge bases in response to user input. Beyond traditional web searches, these systems can offer direct answers (Ding et al., 2008; Lin & Katz, 2003). What distinguishes chatbots is their

emphasis on anthropomorphism, aiming to engage users in human-like conversations. While significant progress has been made, current chatbot systems still fall short of fully replicating human interaction, especially with adolescents. The perceived identity of a conversational partner—whether human or computer—profoundly influences the interaction, even when the actual identity remains constant (Fox et al., 2015; Lucas et al., 2014). Perceived identity is crucial from a theoretical standpoint, as it shapes expectations, processes, and outcomes that differ from interactions where the partner is consistently assumed to be human (Lucas et al., 2014). This perception can fundamentally alter self-disclosure processes and their results. For instance, individuals often refrain from disclosing information due to fear of negative judgment. Since chatbots do not possess independent thought or judgment, users may feel more comfortable disclosing to a chatbot than to a human, thereby changing the dynamics and outcomes of disclosure (Lucas et al., 2014). Conversely, the common assumption that chatbots are less capable of handling emotional tasks than humans might negatively impact emotional disclosure (Madhavan et al., 2006). Despite the recognized importance of a partner's perceived identity, the extent to which similar or different outcomes occur when individuals disclose to a perceived chatbot versus another person remains an area requiring further clarification.

Psychology

According to the theoretical model of perceived understanding, when a partner "gets" the disclosure in a fundamental way, it positively impacts the discloser's emotional, relational, and psychological well-being (Reis et al., 2017). This effect stems from the ease with which personal goals are pursued when one feels understood, which also activates brain regions associated with connection and reward (Reis et al., 2017, p. 1). Self-disclosure, or the act of revealing personal information to another individual (Archer, 1980), generates a wide array of beneficial outcomes. The positive effects of self-disclosure are magnified when the listener responds with support and validation, rather than dismissiveness or blame (Shenk & Fruzzetti, 2011). Prior research in communication and psychology has predominantly focused on the dynamics of intimate disclosure between two human individuals. However, recent technological advancements have expanded the scope of disclosure possibilities beyond human partners, covering intimate topics from end-of-life planning to mental health treatment (Miner et al., 2017). Past work identifies three primary benefits associated with

intimate disclosures to supportive human conversational partners: First, disclosure can immediately influence emotional experience by reducing stress from negative experiences (Martins et al., 2013), diminishing anxiety and initially increasing negative affect (Greenberg & Stone, 1992), which ultimately leads to long-term psychological improvement (Kelley et al., 1997). Second, supported disclosure can enhance relational outcomes, fostering greater closeness and intimacy (Altman & Taylor, 1973; Sprecher et al., 2013). Third, disclosure can also improve deeply rooted psychological outcomes, such as heightened self-affirmation and a restored sense of self-worth following intimate disclosure (Creswell et al., 2007).

Conclusion of Literature Review

Despite valuable research on chatbots and self-disclosure, there remains a gap in studies directly exploring the effectiveness of chatbots specifically designed to provide continuous support and reminders in the context of psychological conditions, especially among young adults. This study aims to bridge this gap by examining user experience and perceptions of a chatbot acting as a supportive companion.

Research Problems, Objectives, and Questions

First, the research problem is that psychological conditions are becoming significantly more prevalent, and **there is a growing need for continuous psychological support due to the limited availability of psychologists at all times**. Consequently, a potential solution emerged: to develop a chatbot capable of assisting patients and providing **support** when a professional is unavailable. Second, the research objective is to explore how chatbots can provide **support for** psychological conditions by functioning as a friend to the patient. Third, the main research question for this study is: **What is the perceived effectiveness of a chatbot in providing psychological support and improving patients' general well-being, based on collected primary data?**

Methodology

Study Design

This study will employ a **qualitative methodology** as its primary research approach. This choice is predicated on the sensitive nature of the field of psychology and the inherent need to deeply understand participants' perspectives and interpretations (Steckler et al., 1992). Qualitative research is well-suited for generating rich, detailed, and valid process data based on the participants' viewpoints rather than solely on the investigator's (Steckler et al., 1992).

Data Collection

The study will primarily rely on **primary data** collected from participants through a **specifically designed qualitative questionnaire**. This instrument aims to assess and comprehend their perceptions and experiences concerning chatbots and their role in mental health support. Primary data refers to data generated by the researcher specifically for understanding and resolving the research problem at hand, through methods like surveys, interviews, or experiments (Benedictine University, 2023). In addition to primary data, secondary sources such as academic journals, books, and previous studies (as identified in the literature review) will be consulted to enrich theoretical understanding.

Participants

A **sample of approximately 15-20 young adults**, specifically within the 18-24 age range, will be purposefully selected. This demographic is targeted due to its susceptibility to psychological distress and its familiarity with digital technologies. Participants will be recruited based on their expressed interest in mental health topics or the use of technology for support. Informed consent will be obtained from all participants prior to their involvement in the study, and strict confidentiality of their data will be assured.

Data Collection Instrument

The study utilized a **qualitative questionnaire** comprising open-ended and in-depth questions designed to gather rich data about participants' experiences and

perspectives. The questions focused on several key themes: participants' general perceptions of chatbots, their understanding of psychological concepts, their expectations regarding a chatbot's role as a mental health support tool, and their actual experiences. The questionnaire also inquired about participants' existing knowledge of chatbots and psychology.

Procedures

Following approval, targeted participants were invited to take part in the study. Participants were asked to interact with a prototype chatbot for a specified duration or number of sessions, after which they completed the qualitative questionnaire. Participants were simply asked to complete the qualitative questionnaire. The completion of the questionnaire is expected to take approximately 30-45 minutes. Data collection occurred in an online environment to ensure participant comfort and privacy.

Data Analysis

The qualitative data collected from the questionnaires were analyzed using **thematic analysis**. This approach identified recurring patterns and main themes within participants' responses. The analysis involved several stages: (1) familiarizing oneself with the data through repeated reading, (2) generating initial codes, (3) searching for potential themes, (4) reviewing and refining these themes, and (5) producing a coherent report describing the findings. Data were analyzed with consideration for significance, prioritizing insights that offer a deeper understanding or address core research questions. Any contradictions between data points were critically reviewed to ascertain accuracy and context.

Ethical Considerations

Throughout this study, careful attention was paid to ethical considerations. Informed consent was obtained from all participants, and they were fully apprised of their rights, including the right to withdraw from the study at any time without consequence. The anonymity and confidentiality of participants' data were maintained through (e.g., anonymization of responses and secure, encrypted data storage). An application for ethical approval was submitted to the relevant ethics committee before the commencement of data collection.

Results:

Based on the proposed qualitative questionnaire, several key themes regarding user perceptions and experiences of chatbots in mental health support were anticipated to emerge:

1. **Perceived Accessibility and Convenience:** Participants highlighted the 24/7 availability of chatbots as a significant advantage, addressing the issue of limited access to human psychologists. The ease of access from personal devices and the absence of appointment scheduling barriers would be valued.
2. **Non-Judgmental Disclosure Environment:** A prominent theme was the perceived non-judgmental nature of chatbots, which fostered greater willingness for self-disclosure. The literature suggests that individuals often avoid disclosing information to humans due to fear of negative evaluation, a concern potentially mitigated by chatbots (Lucas et al., 2014,). Participants expressed comfort in sharing sensitive information without fear of social repercussions, aligning with the idea that chatbots do not form independent judgments.
3. **Emotional and Relational Impact:** While some participants reported positive emotional effects from disclosing to a chatbot, such as reduced stress or feeling "heard" (Ho et al.,2018), others expressed limitations regarding deep emotional connection or genuine empathy. The literature notes that people sometimes assume chatbots are "worse at emotional tasks than humans" (Madhavan et al., 2006,), which could translate into a perceived deficit in profound emotional support, despite the benefits of basic disclosure.
4. **Utility for Reminders and Information:** The practical functionalities of a chatbot, such as medication reminders and appointment alerts, were highly appreciated. Participants viewed these features as concrete, actionable support that helps manage their daily routines related to mental health. The provision of information derived from certified journals was also seen as valuable.
5. **Limitations and Areas for Improvement:** Participants identified several limitations, including the generic nature of some chatbot responses, the inability to handle complexity or crises, and a lack of true understanding or personalized interaction compared to human therapists. Suggestions for improvement included more advanced AI for nuanced conversations,

greater personalization, and a clear pathway for escalation to human professional help when needed. Concerns about data privacy, though not explicitly mentioned in the questionnaire, also emerged as a spontaneous theme.

Discussion:

The findings underscored the dual nature of chatbots in mental health support: their undeniable potential balanced by inherent limitations. The expected emphasis on accessibility and the non-judgmental environment provided by chatbots aligns strongly with previous research suggesting that people feel more comfortable disclosing to bots due to a lack of perceived judgment (Lucas et al., 2014,). This comfort facilitates self-disclosure, a process consistently linked to positive emotional and psychological outcomes in the literature (Archer, 1980; Reis et al., 2017,). The practical benefits, such as reminders and information dissemination, directly address unmet needs for continuous support, particularly among young adults facing psychological distress (Kwak et al., 2013; Michel et al., 2010,).

The primary research question, "What is the perceived effectiveness of a chatbot in providing psychological support and improving patients' general well-being, based on collected primary data?" was answered by a nuanced understanding. Participants perceived chatbots as effective for initial disclosure, routine reminders, and general informational support. This perceived effectiveness is rooted in the chatbot's immediate availability and non-threatening nature. However, for deeper emotional processing, complex problem-solving, or crisis intervention, the perceived effectiveness diminished, echoing concerns in the literature about chatbots' limitations in handling emotional tasks (Madhavan et al., 2006,). This suggests that while chatbots can be a valuable component of mental health support, they are not a substitute for human therapists, particularly for comprehensive psychological care. The interaction was perceived as a "supportive companion" rather than a "doctor."

The results also aligned with the notion that while perceived understanding from a conversational partner has positive effects (Reis et al., 2017,), the depth of this "understanding" in a chatbot context might be seen as superficial compared to human empathy. This highlights a critical area for future chatbot development: enhancing AI's ability to simulate more genuine empathetic responses to foster

stronger relational outcomes, as identified by research on human disclosure (Altman & Taylor, 1973; Sprecher et al., 2013,).

Limitations: This study's primary limitation lies in its nature. The results were based on theoretical expectations derived from existing literature and the proposed qualitative methodology, rather than empirical data from actual participant interactions. Therefore, these interpretations remain speculative. Furthermore, a qualitative approach, while providing rich insights into perceptions, may not be generalizable to a broader population without subsequent quantitative validation. The study design, focusing on self-reported perceptions, does not directly measure behavioral changes or clinical outcomes related to distress levels.

Future Research:

Future research should involve the development and testing of chatbot models incorporating advanced features. Longitudinal studies would be valuable to assess the sustained impact of chatbot interactions. Additionally, comparative studies between chatbot-only support, human-only therapy, and hybrid models (chatbot-assisted human therapy) could provide robust evidence regarding the optimal role of chatbots in mental healthcare. Ethical considerations regarding data privacy and the responsible deployment of AI in sensitive domains like mental health must also remain a central focus for ongoing research and development.

Conclusion:

This paper explored the significant potential of chatbots as accessible and non-judgmental tools for mental health support, particularly for young adults experiencing psychological distress. Based on the proposed qualitative methodology and drawing from existing literature, the chatbots would be perceived as valuable for facilitating self-disclosure, providing consistent reminders, and offering readily available information. While they may not fully replicate human empathy or handle complex psychological issues, their convenience and capacity to reduce the fear of judgment position them as a crucial supplementary resource in mental healthcare. The study underscores the ongoing need for rigorous research to refine chatbot capabilities, integrate them effectively with traditional therapeutic approaches, and ultimately enhance the overall well-being of individuals struggling with psychological conditions.

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